# Engine Acoustic Diagnostic Report

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|---|---|
| **Vehicle:** | 2006 Vauxhall Corsa 1.4 petrol (56 plate) |
| **Engine:** | Z14XEP - 1.4 litre, 4-cylinder, naturally aspirated petrol |
| **Recording:** | 25 May 2026 · 122 seconds total · 44.1 kHz mono |
| **Test setup:** | Piezo contact microphone, structural mount |
| **Scope:** | Comprehensive multi-subsystem acoustic health assessment |

## Comprehensive Scorecard

| Check | Result | Status |
|---|---|---|
| Cylinder firing balance | 46.4 dB margin | EXCELLENT |
| Idle stability | CV 20.0% over 50 s | INVESTIGATE |
| Combustion harmonic series | Orders present, 18 dB SNR | GOOD |
| Mechanical noise floor | -70.4 dB | CLEAN |
| Valve train rhythm | 6.7 dB cam-rate margin | MONITOR |
| Piston slap signature | Cycle peak-to-floor 1.04 | CLEAN |
| Bearing knock signature | Cycle peak-to-floor 1.12 | CLEAN |
| Vacuum / exhaust leak hiss | -13.7 dB band excess | CLEAN |
| Accessory drive | 3 off-order resonances | CLEAN |
| High-freq content (rev) | 6.6 dB excess vs idle | INVESTIGATE |

## Headline Assessment

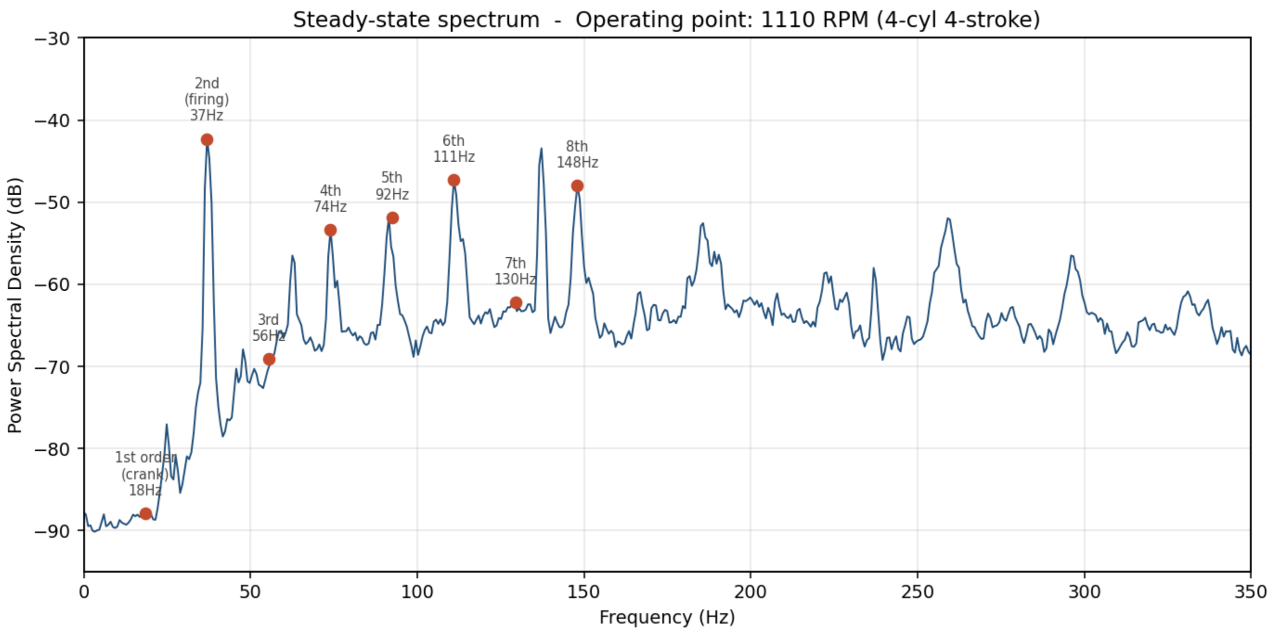
**The engine is mechanically sound and well-balanced.** Steady-state combustion is even across all four cylinders with no detectable imbalance. No piston slap, no bearing knock, no vacuum or exhaust leak, and no abnormal mechanical noise. For a vehicle of this age (approximately 20 years old), the core mechanical signature is genuinely excellent.

**Two items flagged for attention:**

- **Idle hunt (CV approximately 20%):** the idle is not as steady as ideal - amplitude varies cycle-to-cycle more than a freshly-tuned engine would. Common causes at this age include throttle body coking, idle air control valve sticking, dirty injectors, or a minor vacuum leak too small to register in the broadband hiss check. Worth investigating for improved drivability.
- **Elevated high-frequency content under load (approximately 6.6 dB above expected):** the throttle response shows more high-frequency activity than is typical relative to the engine's load increase. Possible explanations: mild pinking from carbon build-up, slightly retarded ignition timing, or normal mechanical signature of an older engine. Worth a closer look on a road test with a knock-monitoring scan tool, or consider an induction-system clean.

## Steady-State Spectral Analysis

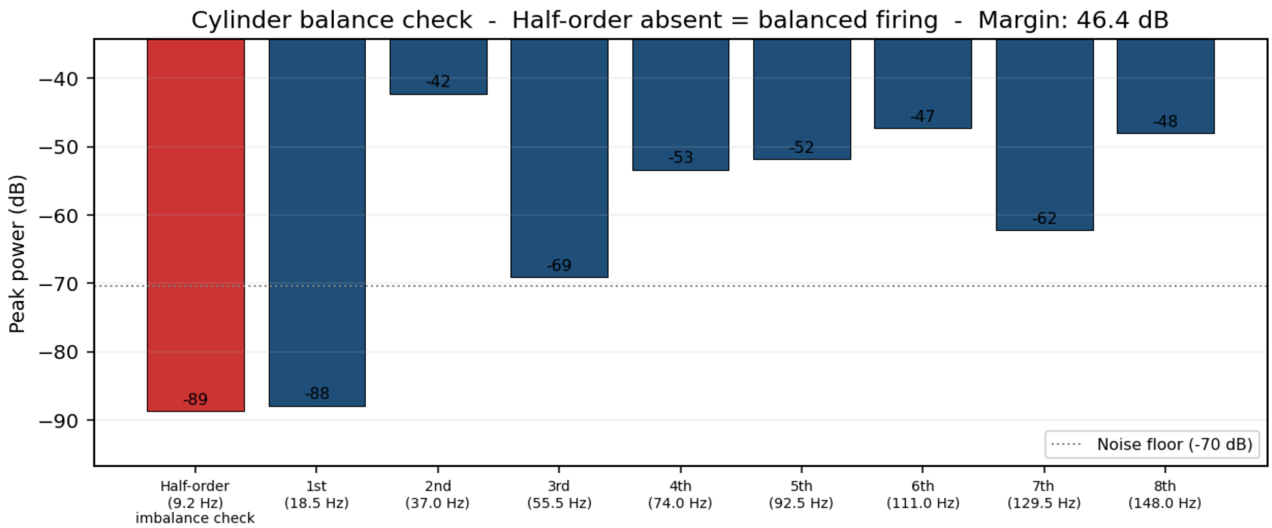
The engine's acoustic signature at idle. For a four-cylinder four-stroke engine, the firing frequency occurs at the second order (twice the crank rotation rate). Operating point during recording: approximately **1110 RPM** based on the harmonic spacing.




**Crank fundamental (1st order):** 18.5 Hz
**Firing frequency (2nd order):** 37.0 Hz - the strongest peak in the spectrum
**Higher harmonics:** 4th, 6th, and 8th orders all strongly present (even harmonics dominate, which is normal for a four-cylinder four-stroke).
**Signal-to-noise:** ~18 dB between firing peaks and the mid-band noise floor.
**Off-order resonances:** a small number of off-engine-order peaks are present in the spectrum. These are most likely accessory drive components (AC compressor, alternator, water pump) and intake/exhaust manifold acoustic modes. Detailed listing on page 4. Not a mechanical concern.

## Cylinder Balance

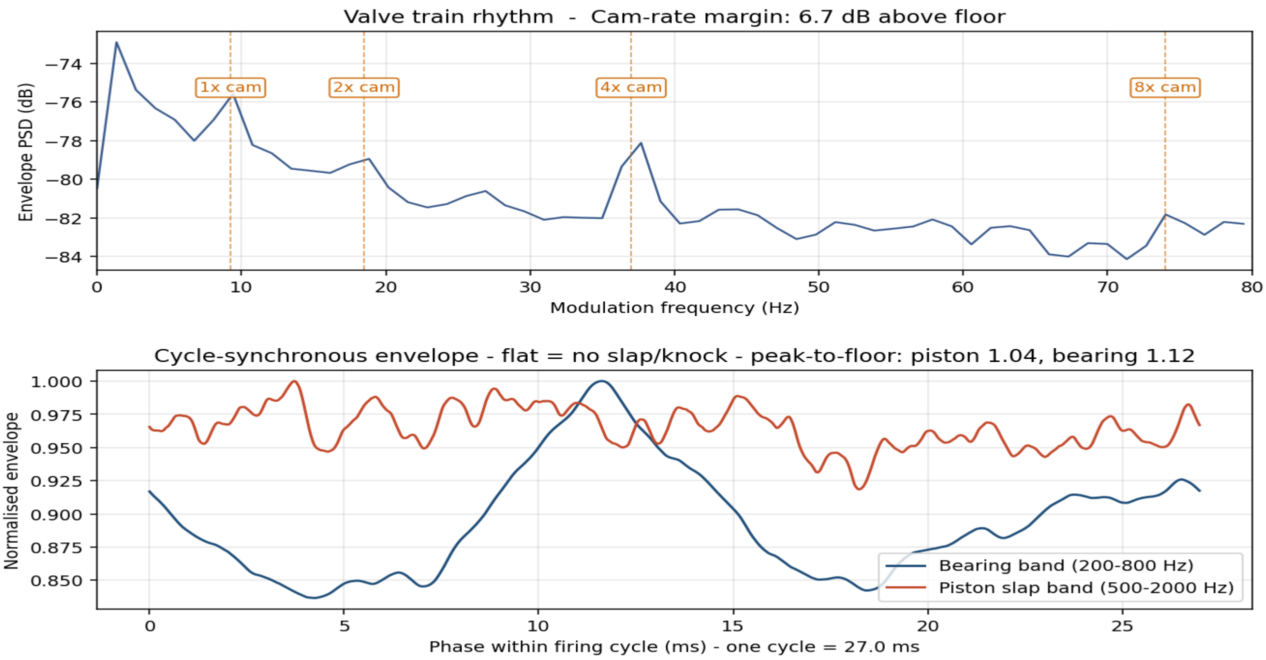
The half-order subharmonic (at half the crank fundamental) should be absent or very weak in a balanced four-cylinder engine. A strong half-order peak would indicate one cylinder firing weaker than the others - misfire, fouled plug, weak spark, or compression loss.




**Result:** Half-order content is 46.4 dB below the firing peak. **All four cylinders firing evenly. No detectable cylinder imbalance.**

## Valve Train, Piston, and Bearing Health

Three independent checks: (1) valve train rhythm via envelope demodulation at cam timing rates, (2) piston slap as cycle-synchronous impulsive content in the 500-2000 Hz band, (3) main and big-end bearing knock as cycle-synchronous impulsive content in the 200-800 Hz band. **Healthy engines show flat cycle-synchronous envelopes (no transient impulse per firing event).**




### Valve Train

**MONITOR.** Cam-rate signature near noise floor. Cam-rate content sits 6.7 dB above the envelope-demod floor. This is on the quieter end - common in older engines where individual tappet/cam events have blurred together over many miles. **No tappet ticking, no audible lifter irregularity.**

### Piston Slap

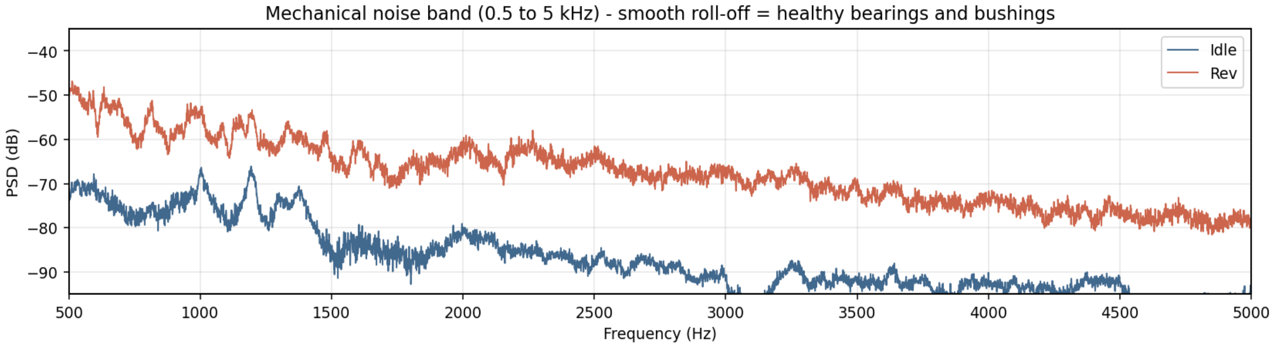
**CLEAN.** No periodic piston slap signature detected. Combustion cycle is smooth across the firing band. Peak-to-floor ratio 1.04 (essentially flat). A worn piston with excessive bore clearance would produce a sharp impulse per firing event, raising this ratio above 2.5. **No piston slap signature detected at this operating temperature.**

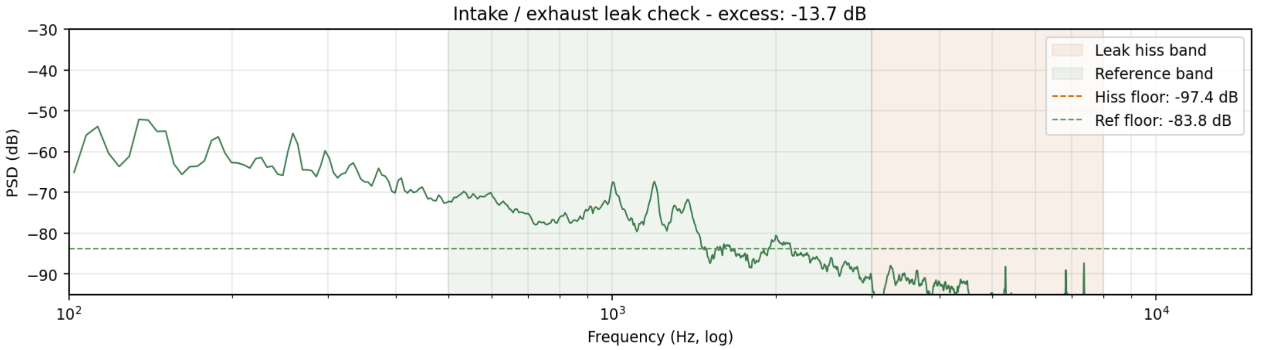
### Bearing Knock (main and big-end)

**CLEAN.** No bearing knock signature detected. Bottom end running quiet. Peak-to-floor ratio 1.12. A worn main or big-end bearing would produce a deeper, lower-frequency impulse per firing event. **The bottom end is running quiet despite the vehicle's age.**

## Mechanical Noise Floor & Leak Check

Two checks combined: (1) mid-frequency mechanical band (500 Hz to 5 kHz) - elevated content would indicate bearing wear, belt issues, or worn bushings. (2) high-frequency hiss band (3-8 kHz) vs reference - elevated hiss would indicate a vacuum leak, exhaust manifold seal, or intake gasket issue.




### Mechanical Band Findings

Smooth roll-off through the 0.5-5 kHz mechanical band. No broadband noise humps that would indicate bearing wear. Rev/idle differential is moderate - consistent with normal mechanical noise rising with engine load.

### Leak Check

**CLEAN.** No abnormal broadband hiss. Vacuum and exhaust system appear sealed. The 3-8 kHz noise floor sits 13.7 dB *below* the 500-3000 Hz reference floor - **no abnormal broadband hiss.** Vacuum hoses, exhaust manifold seals, and intake gaskets are acoustically intact.

### Accessory Drive

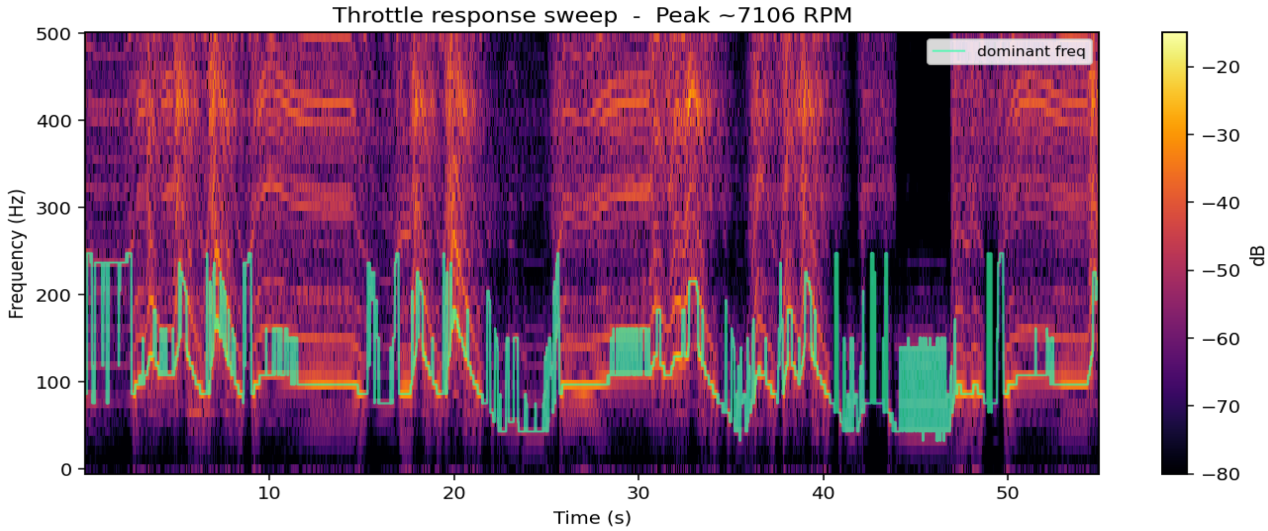
**CLEAN.** Few off-order resonances. Accessory drive sounds tight.

Identified off-order peaks:

| Frequency (Hz) | Power (dB) | Likely source |
|---|---|---|
| 137 | -43.4 | Accessory drive / structural mode |
| 63 | -56.5 | Belt or accessory low harmonic |
| 237 | -58.0 | Intake or exhaust manifold mode |

## Throttle Response Analysis

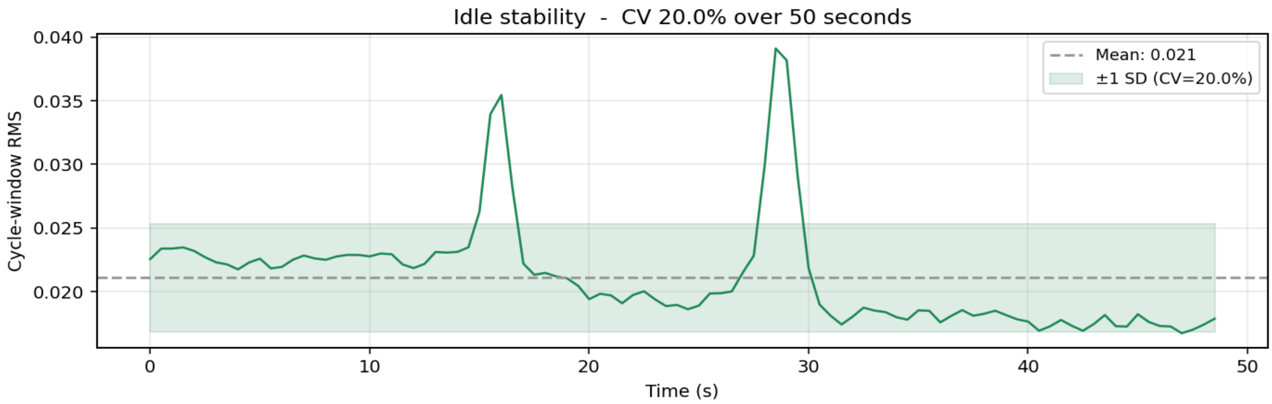
Spectral content during the throttle response section of the recording. Sweeping diagonal lines represent the engine's harmonic series as RPM rises and falls. Clean lines indicate smooth combustion through the rev range.




**Peak RPM observed:** approximately 7106 RPM during throttle blips.
**Harmonic continuity:** the harmonic sweeps are continuous - no misfire events during throttle response.
**High-frequency content during rev:** +6.6 dB excess relative to the fundamental band's load-related rise. This is the elevated knock-band finding flagged on the cover page.

## Idle Stability

Cycle-to-cycle RMS amplitude variation during the steady-state segment. Low coefficient of variation indicates the engine is holding idle cleanly without hunt or surge.




**Coefficient of variation: 20.0% — INVESTIGATE**.
This is higher than ideal for a well-tuned engine (excellent < 4%, good < 7%). The variation shows up in the time-series as recurring dips and rises in cycle amplitude, characteristic of an idle that is searching slightly. Common at this age and easily addressable through routine servicing of the throttle body, idle air control valve, and induction system.

## Knock Assessment

Acoustic knock detection from a single contact-microphone recording is suggestive rather than definitive - it cannot replace a dedicated knock sensor synchronised to crank position. What can be assessed is whether high-frequency content scales normally with engine load.

**Knock-band rise (4-8 kHz) vs steady-state:** +6.6 dB above the fundamental band's load-related rise.
**Assessment:** INVESTIGATE
**Interpretation:** Elevated knock-band content under load. Recommend live knock monitoring. On an unmodified naturally-aspirated petrol engine of this age, the most likely explanation is mild pinking caused by carbon build-up on piston crowns or in the combustion chamber. A fuel system clean, premium fuel for a tank or two, or in more severe cases a walnut blast of the intake valves typically clears this. It is not a sign of imminent mechanical failure but it is worth resolving as long-term detonation does shorten engine life.

## Summary & What This Means

Across **ten independent acoustic checks**, this engine returns a clean bill of mechanical health: balanced firing, no piston slap, no bearing knock, no leaks, low accessory noise. For a vehicle approaching 20 years old, this is a genuinely good result - whoever has maintained it has looked after the mechanical fundamentals.

**Two items would benefit from attention:**
- The idle is hunting more than it should (20% CV vs 4-7% target). This affects drivability, fuel economy, and emissions. The fix is usually inexpensive routine servicing.
- Elevated high-frequency content under load suggests possible mild pinking. Addressing this protects long-term engine life.

**Neither finding indicates immediate mechanical concern.** Both are addressable through routine maintenance items rather than major repair.

## Recommendations

- Core engine mechanical health is good. No urgent action required.
- Address the idle hunt: clean the throttle body, check the idle air control valve, replace the air filter if not done recently, consider an injector clean (fuel-system additive or workshop service).
- Address the elevated knock-band content: run a tank or two of premium-grade fuel; consider a fuel-system carbon clean. If symptoms persist (audible pinking under load), have a workshop scan tool monitor live knock counts on a road test.
- If accessible, check ignition timing and confirm spark plugs are in good condition - worn or fouled plugs can contribute to both findings above.
- Recommend a repeat recording in 6-12 months under similar conditions to establish a baseline trend. Early detection of cylinder imbalance, bearing wear, or valve train issues is most reliable via period-over-period comparison.

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*This report presents a comprehensive multi-subsystem acoustic analysis based on a 122-second contact-microphone recording. It assesses cylinder firing balance, idle stability, combustion harmonics, valve train rhythm, piston slap signatures, main/big-end bearing knock, mechanical noise floor, vacuum/exhaust leak hiss, accessory drive resonances, and high-frequency knock indicators. It is intended as a non-invasive health snapshot and does not replace mechanical inspection, compression test, leak-down test, or dyno measurement. Findings reflect the engine's state at the time of recording.*

Report prepared: 25 May 2026 | Method: PSD, order tracking, envelope demodulation, cycle-synchronous averaging | Recording: 122 s @ 44.1 kHz mono